

Homework 6

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```
library(bio3d)
```

Comments from rubric

Inputs to the function: The protein ID from the Protein Data Bank (PDB)

What the function does and how to use it: This function uses the protein ID to analyze proteins by trimming them to a specific chain (A) and atom/element type (CA), and reading their b-factor values. This is used to generate a plot from any PDB ID, with the specific characteristics wanted. This function is used by using the function `pdb.plot` and using the PDB ID as the input.

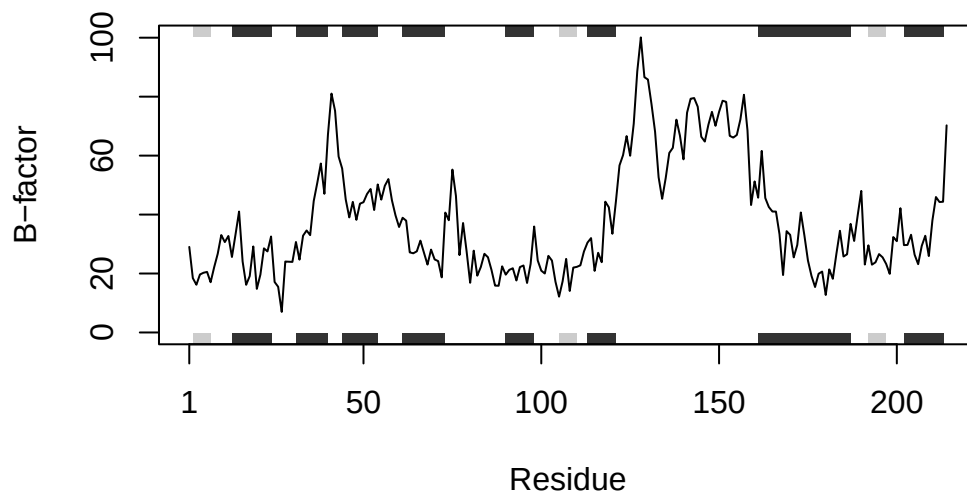
Output of the function: The output of the function is a b-factor plot as well as the b-factor data in the form of a vector.

```
pdb.plot <- function(pdb_id) {  
  
  structure <- read.pdb(pdb_id)  
  
  structure.trim <- trim.pdb(structure, chain = "A", elety = "CA")  
  
  b <- structure.trim$atom$b  
  
  plotb3(b, sse = structure.trim, typ = "l", ylab = "B-factor")  
  
  return(b)  
}
```

Function works and correct output is produced

```
pdb.plot("4AKE")
```

Note: Accessing on-line PDB file



[1]	29.02	18.44	16.20	19.67	20.26	20.55	17.05	22.13	26.71	33.05
[11]	30.66	32.73	25.61	33.19	41.03	24.09	16.18	19.14	29.19	14.79
[21]	19.63	28.54	27.49	32.56	17.13	15.50	6.98	24.07	24.00	23.94
[31]	30.70	24.70	32.84	34.60	33.01	44.60	50.74	57.32	47.04	67.13
[41]	81.04	75.20	59.68	55.63	45.12	39.04	44.31	38.21	43.70	44.19
[51]	47.00	48.67	41.54	50.22	45.07	49.77	52.04	44.82	39.75	35.79
[61]	38.92	37.93	27.18	26.86	27.53	31.16	27.08	23.03	28.12	24.78
[71]	24.22	18.69	40.67	38.08	55.26	46.29	26.25	37.14	27.50	16.86
[81]	27.76	19.27	22.22	26.70	25.52	21.22	15.90	15.84	22.44	19.61
[91]	21.23	21.79	17.64	22.19	22.73	16.80	23.25	35.95	24.42	20.96
[101]	20.00	25.99	24.39	17.19	12.16	17.35	24.97	14.08	22.01	22.26
[111]	22.78	27.47	30.49	32.02	20.90	27.03	23.84	44.37	42.47	33.48
[121]	44.56	56.67	60.18	66.62	59.95	70.81	88.63	100.11	86.60	85.80
[131]	77.48	68.13	52.66	45.34	52.43	60.90	62.64	72.19	66.75	58.73

[141]	74.57	79.29	79.53	76.58	66.40	64.76	70.48	74.84	70.11	74.82
[151]	78.61	78.24	66.70	66.10	67.01	72.28	80.64	68.54	43.23	51.24
[161]	45.72	61.60	45.61	42.57	41.03	41.02	33.34	19.48	34.38	33.11
[171]	25.48	29.68	40.71	32.91	24.41	19.20	15.43	19.93	20.66	12.72
[181]	21.40	18.21	26.68	34.50	25.77	26.52	36.85	31.05	39.84	48.03
[191]	23.04	29.57	23.00	23.80	26.59	25.49	23.25	19.89	32.37	30.97
[201]	42.16	29.64	29.69	33.15	26.38	23.17	29.35	32.80	25.92	38.01
[211]	45.95	44.26	44.35	70.26						